

# Jack He

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## EDUCATION

### Boston University

Boston, MA

*B.A. in Computer Science, Minor in Data Science, GPA: 3.93*

*Expected May 2029*

- **Relevant Coursework:** Computer Systems, Geometric Algorithms, Combinatoric Structures, Advanced Data Structures in Java, Software Engineering Career Prep

## EXPERIENCE

### Founding Software Engineer

Feb. 2026 – Present

#### *Sapling*

*Boston, MA*

- Won Best AI in Education at BU's 2026 CivicHacks hackathon for an open-source AI study platform that supports three learning modes (Socratic, expository, teach-back) and tracks student mastery through a real-time D3.js knowledge graph
- Architected the full stack on Next.js and FastAPI, containerized all services with Podman, and set up continuous deployment via Railway so the team could ship features without manual intervention
- Built real-time collaborative study rooms using WebSockets for live chat, partner-based matchmaking, and side-by-side knowledge graph comparison across room members
- Delivered 8 end-to-end features (tutoring, quizzing, flashcards, study rooms) across the frontend and backend, owning each from design through deployment

### Software Development Intern

Mar. – May 2025

#### *Atlas Sports*

*Boston, MA*

- Built an LLaVA-powered chatbot that analyzes match video history to give hockey players actionable feedback on their gameplay, handling video ingestion and LLM-based summarization end-to-end
- Collaborated with a team of five to build a free, open-access social platform for school sports teams, taking it from concept to functional prototype in under three months

### Microelectronics and Hardware Development Student

Jul. – Aug. 2024

#### *MIT Lincoln Labs Beaver Works Summer Institute*

*Boston, MA*

- Selected as 1 of 41 students nationally for MIT Lincoln Labs BWSI, gaining hands-on experience in IC design, PCB layout, and microchip fabrication over an intensive two-month program

## PROJECTS

### **Waymark** | *Meta Llama 3.1 8B, Python, React, Node.js, HTML/CSS, ChromaDB, RAGAs*

Mar. 2026

- Won Best Overall Track at Red Hat's Open Accelerator hackathon for building a multimodal RAG system ingesting Google Drive and Slack data to answer employee queries, estimated to save 1.8 hours of search time daily across platforms for companies
- Delivered sub-2-second answers with 93.6% accuracy across 847 documents using a hybrid search and reranking pipeline, self-hosted on 2x NVIDIA H100 GPUs

### **Stalk Market** | *OpenAI, Python, JavaScript, HTML/CSS*

Oct. 2025

- Won Best Technical Execution at BU's 2025 Data Science + X hackathon for an interactive stock market simulation game with Geometric Brownian Motion pricing and custom AI trading agents
- Designed the full game loop from market engine to frontend, letting players compete against AI agents that adapt their trading strategies based on real-time price signals

### **Super Artificial Bros.** | *Gemini API, Python, JavaScript, Node.js*

Oct. 2025

- Won Best Use of AI and LLMs at BostonHacks 2025 for an AI-generated Mario game that uses Gemini to create unique levels from existing terrain data and drives NPC dialogue through a relationship algorithm
- Built a contextual memory system that tracks long-term player decisions and dynamically adjusts gameplay, NPC behavior, and level difficulty based on accumulated play history

### **SIGMAS** | *Arduino, Python, C++, OpenCV, TensorFlow, YOLOv8*

Aug. 2024

- Built a home security device on a \$50 Arduino budget with ESP32 cameras, using OpenCV for facial recognition and a custom TensorFlow/YOLOv8 model trained to detect weapons in real time
- Designed the full hardware-software pipeline from camera capture to on-device inference, optimizing model size to run within the ESP32's memory constraints

### **Lightout PRO** | *Arduino, C++*

Aug. 2024

- Placed 3rd in the 2024 IEEE SSCS Arduino Contest for an automated lighting system using ultrasonic, PIR motion, and photoresistors to reduce household energy consumption
- Assembled the full system and published it within one week, writing C++ firmware that coordinates three sensor types into a single energy-saving control loop

## TECHNICAL SKILLS

**Languages:** Java, Python, C, C++, JavaScript, Assembly, SQL, HTML/CSS

**Frameworks and Libraries:** Next.js, FastAPI, Node.js, D3.js, React, OpenCV, TensorFlow, YOLOv8, WebSockets

**Tools and Platforms:** Git/GitHub, Docker/Podman, Linux, Railway, VS Code, Arduino, ESP32, Canva, Adobe Animate